

Public Data Availability Report

Location Intelligence - Exploratory Assessment

Prepared for: DFW pilot / general site-selection use cases

Date: July 6, 2026 (updated with live pilot test results)

Scope: Nine public sources requested for DFW pilot counties (Collin, Dallas, Denton, Rockwall, Tarrant)

Purpose: Confirm what public data exists, API access, cost, granularity, and what actually worked in production pulls.

Executive Summary

A live pilot test on July 6, 2026 confirmed that 8 of 9 requested sources deliver usable data into the pipeline automatically. Only Texas RRC is partial: the public MFT catalog lists per-county API DBF files, but programmatic download requires a browser session (GoDrive/PrimeFaces). Three free API keys power the keyed sources: Census (ACS, PEP, CBP), FRED (building permits), and data.gov (FBI CDE crime). The remaining six sources require no API key - they use public CSV/XLSX downloads or open ArcGIS/Socrata REST endpoints.

Pilot Live Test Results (July 6, 2026)

Connectivity probe: 9/9 OK

Source	API key?	Pilot status	Access method
Census ACS (acs/acs5)	CENSUS_API_KEY	WORKING	api.census.gov REST
Census PEP (pep/charv)	CENSUS_API_KEY	WORKING	api.census.gov REST
FRED building permits	FRED_API_KEY	WORKING	fred.stlouisfed.org REST
FBI CDE crime	DATA_GOV_API_KEY	WORKING	api.usa.gov/crime/fbi/cde/...
	Y		
FHFA HPI	None	WORKING	fhfa.gov annual county XLSX download
Zillow ZHVI	None	WORKING	zillowstatic.com public CSV download
FHWA HPMS traffic	None	WORKING	geo.dot.gov ArcGIS REST (county_code)
BTS NTAD	None	WORKING	data.transportation.gov Socrata 8j5p-a5aj
BLM MLRS leases	None	WORKING	gis.blm.gov ArcGIS spatial query
Texas RRC wells	None	PARTIAL	mft.rrc.texas.gov catalog only; no auto-download

Bonus: Census CBP (County Business Patterns) also works with CENSUS_API_KEY. Legacy FBI SAPI (api.usa.gov/crime/fbi/sapi) returns 404 - use CDE summarized endpoints instead. HPMS geospatial dataset 4dez-3n4e on BTS is non-tabular; use HPMS County Summary 8j5p-a5aj instead.

Sample live values (DFW pilot counties)

county	population	mean_aadt	zillow_zhvi	fhfa_hpi	blm_leases	road_miles
Dallas	2,621,179	50,050	\$312,652	986	97	6,248
Tarrant	2,167,390	40,490	\$325,761	882	321	6,414
Collin	1,163,337	32,962	\$488,699	852	142	3,599
Denton	979,561	32,839	\$444,147	825	325	3,411
Rockwall	123,617	24,644	\$417,027	573	20	532

Outputs: data/raw/*.csv (per source), data/processed/county_features.csv (stitched), exports/county_dashboard.csv (Power BI), data/location_intelligence.db (SQLite). Run: python scripts/probe_sources.py (connectivity) or python -m src.run_pipeline (full pull).

Category	API?	Cost	Best granularity	Pilot result
Census population	Yes (1 key)	Free	Block group / tract	WORKING
Traffic	Partial (no key)	Free	Road segment / county	WORKING
Crime	Yes (data.gov)	Free	State rates (agency TBD)	WORKING*
Housing prices	Partial (no key)	Free	County / tract	WORKING
Mineral rights	Limited (no key)	Free	Lease bbox / bulk	PARTIAL

*Crime: live state-level rates via FBI CDE; county-level requires agency crosswalk (not yet built).

1. Census Data - Populations

Primary public sources

Source	Access	API	Cost	Granularity
ACS 5-Year	api.census.gov	Yes	Free	Nation -> block group
ACS 1-Year	api.census.gov	Yes	Free	65k+ pop areas
Decennial Census	api.census.gov	Yes	Free	Block
Population Estimates (PEP)	api.census.gov/pep/charv	Yes	Free	County
CBP (business)	api.census.gov	Yes	Free	County, ZIP

API key required (free): https://api.census.gov/data/key_signup.html. Activation email link may be required; keys can take hours to become active. Rate limits apply (~500 queries/day without key historically; higher with registered key).

Recommended variables for site selection

Population (B01001_001E), median household income (B19013_001E), households (B11001_001E), median home value (B25077_001E), commute time (B08303), age distribution, education, renter/owner mix.

Sample API response (ACS 5-Year, Dallas County, TX)

```
GET https://api.census.gov/data/2024/acs/acs5
?get=NAME,B01001_001E,B19013_001E
&for=county:113&in=state:48&key=YOUR_KEY

[
  ["NAME", "B01001_001E", "B19013_001E", "state", "county"],
  ["Dallas County, Texas", "2621179", "76547", "48", "113"]
]
```

Sample county-level record (DFW pilot - live pull)

county_fips	county_name	population	median_income	households
48085	Collin County, Texas	1,163,337	121,600	416,646
48113	Dallas County, Texas	2,621,179	76,547	982,737
48439	Tarrant County, Texas	2,167,390	84,207	782,419

Verdict: WORKING in pilot. ACS + PEP + CBP all confirmed with CENSUS_API_KEY. Dallas PEP 2023: 2,606,358.

2. Traffic Data

Primary public sources

Source	Access	API	Cost	Granularity
FHWA HPMS	geo.dot.gov ArcGIS REST	Yes (no key)	Free	Road segment
BTS NTAD County Summary	data.transportation.gov/8j5p-a5aj	Yes (no key)	Free	County miles
BTS NTAD geospatial	data.transportation.gov/4dez-3n4e	No (non-tabular)	Free	Road segment GIS
TxDOT Roadway Inventory	txdot.gov	Bulk / reports	Free	County, road
BTS TranStats	transtats.bts.gov	Download	Free	Airport-centric
INRIX / HERE	Commercial	Paid API	Paid	Real-time segment

Pilot fix: use lowercase field `county_code` (not `COUNTY_CODE`) on geo.dot.gov Texas FeatureServer. Example: where=`county_code=113 AND aadt>0` returns live segments. County aggregation no longer requires shapefile download for the pilot - REST pagination is sufficient. BTS county summary dataset 8j5p-a5aj provides systemlength by stateid + countyid (3-digit FIPS suffix).

Sample HPMS segment record

route_id	county_code	aadt	func_class	lanes	speed_limit
IH0030-KG	48113	245000	11 (Interstate)	8	65
US0075-LX	48085	42800	14 (Principal Arterial)	4	55
FM0543	48397	12400	16 (Collector)	2	45

Sample county aggregation (live pilot pull)

county_fips	mean_aadt	max_aadt	p90_aadt	segment_count	ntad_system_miles
48113	50,050	267,131	148,977	12,787	6,248
48085	32,962	231,767	97,073	4,165	3,599
48439	40,490	213,306	124,035	10,870	6,414

Verdict: WORKING in pilot. FHWA ArcGIS + BTS NTAD county summary both return live county data without API keys.

3. Crime Data

Primary public sources

Source	Access	API	Cost	Granularity
FBI Crime Data Explorer	cde.ucr.cjis.gov	Yes	Free	Agency, state, national
FBI SAPI (legacy)	api.usa.gov/crime/fbi/sapi	Unstable	Free	Agency, state
BJS/NIBRS	bjs.ojp.gov	Bulk	Free	Agency / incident
State UCR portals	Varies	Varies	Free	Agency / city
City open data	data.gov / city portals	Often yes	Free	Precinct / tract

Crime is reported by ~18,000 law enforcement agencies voluntarily. County-level statistics are not published uniformly - you must map agency ORI codes to counties and roll up, or use state/national estimates. NIBRS provides incident-level detail (location type, offense, victim demographics) but coverage is incomplete. API key: free via api.data.gov.

Sample CDE API response (live pilot - Texas violent crime rates)

```
GET https://api.usa.gov/crime/fbi/cde/summarized/state/TX/violent-crime
?api_key=DATA_GOV_KEY&from=01-2023&to=12-2023
```

```
{"offenses":{"rates":{"Texas Offenses":{"01-2023":35.24,"02-2023":30.97,...}}}}
```

Sample state estimate record (applied to all pilot counties)

county_fips	violent_rate_per_100k	property_rate	crime_source
48113	34.68	189.39	FBI CDE summarized (state monthly rates)
48085	34.68	189.39	FBI CDE summarized (state monthly rates)

Caveat: Legacy api.usa.gov/crime/fbi/sapi returns 404 in pilot tests. CDE summarized endpoints work with DATA_GOV_API_KEY.
Verdict: WORKING for state-level rates; county stitching still requires agency ORI crosswalk.

4. Housing Prices

Primary public sources

Source	Access	API	Cost	Granularity
FHFA House Price Index	fhfa.gov/hpi/download/annual/hpi_at_county.csv	Bulk XLSX	Free	County (annual)
FRED (FHFA mirror)	fred.stlouisfed.org	Yes	Free	National, MSA, some county
Census ACS	api.census.gov	Yes	Free	Tract+ (median value/rent)
Zillow ZHVI	zillow.com/research/data	CSV download	Free*	ZIP, county, metro
Redfin Data Center	redfin.com/news/data-center	CSV download	Free*	ZIP, county, metro
CoreLogic / CoStar	Commercial	Paid	Paid	Parcel / property

*Zillow/Redfin free for research with attribution; commercial redistribution restricted. FHFA HPI is a repeat-sales index (appreciation), not a median sale price. ACS median home value is survey-based. For site selection, combining FHFA (trends) + ACS (levels) + building permits (growth) is a strong free stack.

Sample FHFA HPI record (annual county XLSX - live pilot)

county_fips	county_name	fhfa_hpi_year	fhfa_hpi_index_nsa
48113	Dallas	2025	986.07
48085	Collin	2025	851.85
48439	Tarrant	2025	882.23

Sample Zillow ZHVI record (live pilot)

county_fips	zillow_region_name	zillow_zhvi	zillow_zhvi_month
48085	Collin County	\$488,699	2026-05-31
48113	Dallas County	\$312,652	2026-05-31

Sample ACS housing record (DFW pilot)

county_fips	median_home_value	households	building_permits_2025
48085	475,600	416,646	19,082
48113	303,000	982,737	12,691

Sample FRED API call (building permits - growth proxy)

```
GET https://api.stlouisfed.org/fred/series/observations
?series_id=BPPRIV048113&api_key=KEY&file_type=json
```

```
{"date":"2025-01-01","value":"12691"} // Dallas County permits
```

Note: FHFA monthly hpi_master.csv no longer contains county rows in 2026; use annual county XLSX instead. Verdict: WORKING in pilot. FHFA county XLSX + Zillow CSV + FRED permits + ACS values - no extra keys beyond Census/FRED.

5. Mineral Rights

Primary public sources

Source	Access	API	Cost	Granularity
TX Railroad Commission (RRC)	rrc.texas.gov	Bulk / queries*	Free	Well / lease / API #
County clerk deed records	County portals	No unified API	Free-fee	Instrument / tract
BLM MLRS (federal)	gis.blm.gov	ArcGIS REST	Free	Federal lease / section
EIA / USGS	eia.gov / usgs.gov	API / bulk	Free	Basin / county stats
Commercial (TexasFile, etc.)	Private	Paid API	Paid	Lease / owner / deed

*RRC explicitly prohibits automated scraping of interactive query systems. Bulk ASCII/JSON datasets are available for download (wellbore, production, permits) but mineral ownership is not a single field - it must be reconstructed from deed chains, leases, assignments, and probate records at the county level. There is no national 'mineral rights API' equivalent to Census.

Sample RRC wellbore record (bulk download)

api_number	county	lease_name	operator	well_type	completion_date
42-113-12345	DALLAS	SMITH UNIT 1	DEVON ENERGY	OIL	2019-04-15
42-085-67890	COLLIN	JOHNSON LEASE	PIONEER NATURAL	GAS	2021-08-22

Sample county deed / mineral instrument (courthouse record)

instrument_type	grantor	grantee	legal_description
Mineral Deed	Smith Family Tr.	ABC Minerals LLC	Sec 12, Blk 33, T2S, Abstract 1234, Ector Co.
OG Lease	Jones Ranch	XYZ Energy Inc.	Sec 8, Blk 41, T1N, Abstract 567, Midland Co.

Sample BLM federal lease count (live pilot - Dallas bbox)

county_fips	blm_lease_count	blm_lease_acres	query_method
48113	97	111,829	GEO_STATE=TX + county bbox envelope
48439	321	152,902	GEO_STATE=TX + county bbox envelope

Texas RRC pilot status

county_fips	rrc_source_file	rrc_well_count	note
48113	api113.dbf	(empty)	Listed on MFT; download needs browser session
48085	api085.dbf	(empty)	Catalog confirmed at mft.rrc.texas.gov public share

Verdict: BLM WORKING via ArcGIS spatial query (no key). RRC PARTIAL - catalog accessible, programmatic DBF download blocked by GoDrive PrimeFaces. Manual workaround: place DBF files in data/raw/rrc_cache/.

6. Integration Status (Post-Pilot)

Implemented and working in location-intelligence pipeline:

- Census ACS + PEP + CBP (CENSUS_API_KEY)
- FRED building permits (FRED_API_KEY)
- FBI CDE crime state rates (DATA_GOV_API_KEY)
- FHFA county HPI XLSX, Zillow ZHVI CSV (no key)
- FHWA HPMS ArcGIS AADT, BTS NTAD county miles (no key)
- BLM MLRS lease counts by county bbox (no key)

Not yet fully automated:

- Texas RRC per-county well counts (MFT download)
- FBI crime at county level (agency ORI crosswalk)
- Mineral ownership from county deed chains (separate project)

API keys required (.env)

```
CENSUS_API_KEY    -> api.census.gov (ACS, PEP, CBP) - free signup
FRED_API_KEY      -> fred.stlouisfed.org (building permits) - free signup
DATA_GOV_API_KEY  -> api.data.gov (FBI CDE crime) - free signup
```

No key needed: FHFA, Zillow, FHWA HPMS, BTS NTAD, BLM MLRS, RRC catalog

Join keys for stitching

Census ACS	GEOID (state+county+tract)	county_fips (5-digit)
Traffic (HPMS)	spatial intersection	county_fips
Crime (FBI)	agency ORI -> county map	state_fips
Housing (FHFA)	county FIPS / tract GEOID	place_name
Mineral (RRC)	API well number	legal description (section/block)

Data gaps to flag in dashboards

- Crime: not all agencies report; county rates may be missing or estimated.
- Traffic: segment-level source data; county mean AADT is an aggregation choice.
- Housing: ACS median value is survey-based; FHFA HPI is an index, not a dollar price.
- Mineral rights: public data shows activity (wells, leases) more reliably than ownership.

Sources

U.S. Census Bureau - api.census.gov, acs/acs5, pep
 FHWA HPMS - fhwa.dot.gov/policyinformation/hpms
 BTS NTAD - data.transportation.gov
 FBI Crime Data Explorer - cde.ucr.cjis.gov
 FHFA HPI - fhfa.gov/data/hpi
 FRED API - fred.stlouisfed.org/docs/api/fred/
 Zillow Research - zillow.com/research/data
 Texas RRC - rrc.texas.gov/resource-center/research
 BLM MLRS - gis.blm.gov